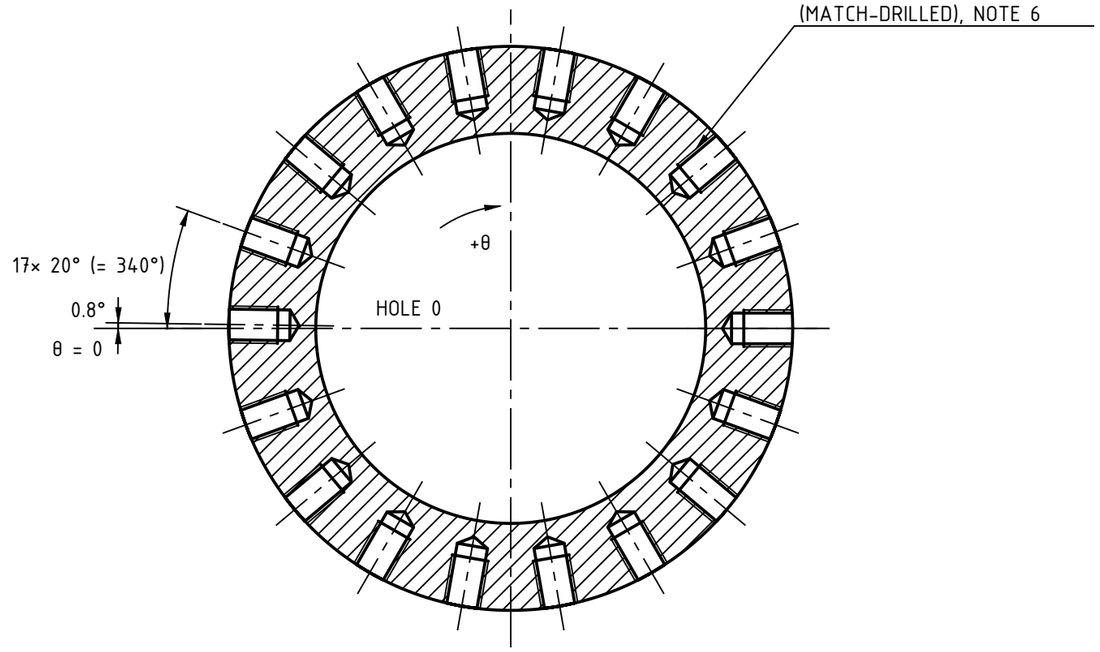
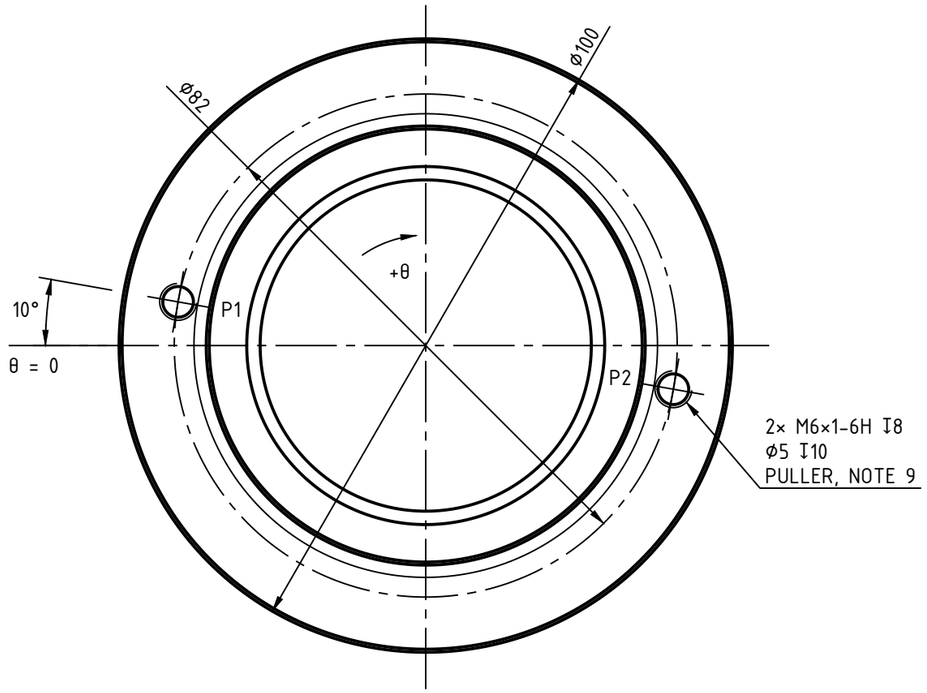
SECTION C–C



VIEW G



NOTES (SHEETS 1 AND 2)

1. Material: stainless steel 304 (1.4301, X5CrNi18-10), ASTM A479 annealed round bar ⌀105 min × 95 min, EN 10204 3.1 certificate. Substitutes (303 included) only with engineering approval.
2. The holder is cut to its own case: land ⌀(A − 0.06) 0/−0.02, case grooves ⌀(A − 5.67) 0/−0.04 (table, sheet 1), A being the honed size of the mating case's aft seal zone, recorded on SRM100-101 (note 4, R4): the smallest reading in the zone, 92.000 ≤ A ≤ 92.070, the zone held to A +0.030/0 (never over 92.10). ⌀91.94 and ⌀86.33 are drawn for A = 92.000. Record A, the cut sizes and the pair serial.
3. Finish-turn the land, both case grooves, the insert bore, the seat face and face B in one setup.
4. Seat root R0.8 ±0.1: smooth blend, no undercut, no step (the bending root of the lip). Glands: bottoms Ra 0.8, walls Ra 1.6, no tool marks across a groove; groove edges R0.1 (0.08 – 0.13).
5. Glands for 2× AS568-237 (case seal, on the land) and 1× AS568-230 (insert seal, in the bore), SRM100-000 items 13 and 14: FKM 90A (Parker V0709-90 or equal), new for every firing. Lubricant: Parker Super O-Lube or DC-55, thin film.
6. 18× M6×1-6H radial holes, made at assembly per SRM100-101 ops 110 – 150, the holder seated and clamped in its case, hole 0 first: tap drill ⌀5 through the case holes, depth stops set from the case OD (the land is covered): full ⌀5 to 14.0 (9.97 from the land), full thread to 12.0 (7.97 from the land); drill point ≥ 2 from the insert bore (2.71 nominal). Bottoming tap through the reamed case holes; M6-6H GO / NO-GO gauge. The SRM100-101 op 130 ream entry cone is permitted (0.5 deep max, ⌀6.40 at the land); full thread is counted below it. Case and holder are then a serialised pair.
7. Clocking: θ is counter-clockwise seen from forward, clockwise seen from aft (SECTION C–C, VIEW G). Until match-drilling θ = 0 is carried by puller P1 (θ 10°, ⌀82): hole 0 (key, θ 0.8°) is 9.2° clockwise of P1 seen from forward, 9.2° counter-clockwise seen from aft. The clocking line at θ = 0 is then scribed across the case end and the flange OD (SRM100-101 ops 120, 140).
8. Mark 'SRM100-102 AFT', the part serial, the pair serial and A (note 2) on the flange OD, centred on hole 0 (θ 0.8°), a 2 mm gap left in the text where the clocking line θ = 0 crosses: 2 mm characters, laser or dot-peen, 0.05 deep max. No marks on the aft body OD ⌀71.58: it carries the temperature labels (sheet 1), irreversible 177 / 204 / 232 / 260 °C strips fitted new at assembly for every firing, at θ 90° and 270°, 15 … 30 from B, clear of the cap tape (SRM100-106 note 6: at θ 0° / 180°, ends ≤ 15 from the aft face).
9. Puller threads P1 / P2 (VIEW G): 2× M6×1-6H ↧8, tap drill ⌀5 ↧10, in the flange aft face on ⌀82 at θ 10° (P1) and 190° (P2); machined in the shop, before match-drilling. For the puller bridge and two draw screws only (tooling). Never lever or knock the holder out of the case.
10. Aft case screws: 18× SRM100-000 item 15, ISO 7380-1 M6×10-010.9, new for every firing. Nickel anti-seize on the threads only, head face dry; 3.0 – 3.5 N·m in the criss-cross sequence of SRM100-000 sheet 2 (3 passes). 6 N·m with anti-seize yields the annealed 304 threads. Never fit an M6×16 here: it bottoms with the head proud.
11. Insert removal: support the holder on its forward face; the push sleeve (tooling, 52.4 ID × 53.8 OD) bears on the insert's aft face through the lip bore. Never push on the lip.
12. Proof test per SRM100-001 with the proof dummy SRM100-120 in place of the insert.
13. Inspect 100 %: land, case grooves and insert groove (⌀ and width), insert bore, seat depth, lip thickness 6 +0.1/0, the geometric tolerances; the glands and the seat root visually at 10×. Record the values with the serial.
14. Deburr and clean: every edge an O-ring passes over is burr-free and smooth (nose chamfer, land, case groove edges, bore lead-in, insert bore, insert groove edges). Tapped-hole entries on the land flush, no raised burr (the land slides in the honed bore). After machining and again after match-drilling: blow out, wash and degrease (IPA); no chips in the glands, threads or bore.
15. After every firing: read and record the temperature labels before extraction; a reading ≥ 232 °C: retire or re-proof (SRM100-001) the aft end. Replace and re-use per SRM100-000 sheet 1 note 13; item 2 is re-used after inspection: clean and check the glands and seat face (no erosion, pitting, scoring), seat root and lip at 10× (no crack), lip thickness and seat depth, all tapped holes with the M6-6H GO / NO-GO gauge. Any finding: quarantine, engineering decides.
16. Surface texture indications per ISO 21920-1 (Ra, ISO 21920-2).

USED ON: SRM100-000 item 2; SRM100-001 item 2

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
304 stainless (1.4301) ASTM A479 annealed bar	1501.2 g	ISO 2768-mK	Ra 3.2	ISO 13715 −0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED – FOR REVIEW	1:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	NOZZLE HOLDER Aft closure – 18× M6 radial – insert bore ⌀63.58

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	–	2026-09-23	SRM100-102	A	2 / 2